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The British Tertiary Igneous Province: Palaeomagnetism and Ages of Dykes, Lundy Island, Bristol Channel

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Summary

Of 66 dykes collected 52 gave stable directions: 44 reversed, 4 normal and 4 intermediate. The mean direction of magnetization was N3·6 °E, +62·7° (down,) equivalent to a pole position of 82·6 °N, 155·0 °E. This is significantly different from the present pole and displays right-handed far-sidedness.

The bulk, at least, of the dykes fall within the apparent span 45–54 My. Yet they display approximately the same small proportion of normal dykes as do the older dyke swarms of the province which were formed during a different polarity period or periods.

1. Introduction

This paper forms part of a palaeomagnetic investigation of the British Tertiary igneous province, particularly of the lavas and dykes.

A collection of 13 dykes was made by Blundell (1957) with the object of ascertaining if the dykes were Tertiary or Permo-Carboniferous; though he did not use magnetic cleaning his results supported the younger age, a conclusion confirmed by radiometric dating (Miller & Fitch 1962; Dodson & Long 1962). The present investigation, using a larger collection and cleaning, is intended to improve our knowledge of the Earth's field in Lower Tertiary times and the relationship of Lundy to the rest of the igneous province.

The island is of Tertiary granite except for the south-west peninsula of slate (Fig. 1), which has been assigned to the Devonian (Dollar 1941). Both rock types are cut by numerous dykes whose predominant direction is approximately NW–SE (Dollar 1941, 1968).

The island appears to form part of a small plutonic complex, the most southerly known in the British Tertiary. A gravity survey (Brooks & Thompson 1973) has shown that the island is on the east side of a positive anomaly about 30 km across and about 20 mgal over the regional value. This is comparable to, but smaller than, those of other centres in the province (Bott & Tuson 1973; Cook 1950; McQuillan & Tuson 1963; Roberts 1970).

The aeromagnetic map shows a strong linear anomaly to the north of Lundy, on a continuation of the Tertiary NW–SE Sticklepath–Lustleigh fault. This direction is characteristic of dyke swarms in the province and the anomaly has been attributed to reversely magnetized dykes (Cornwell 1971). Closer to the island S. Winter (private

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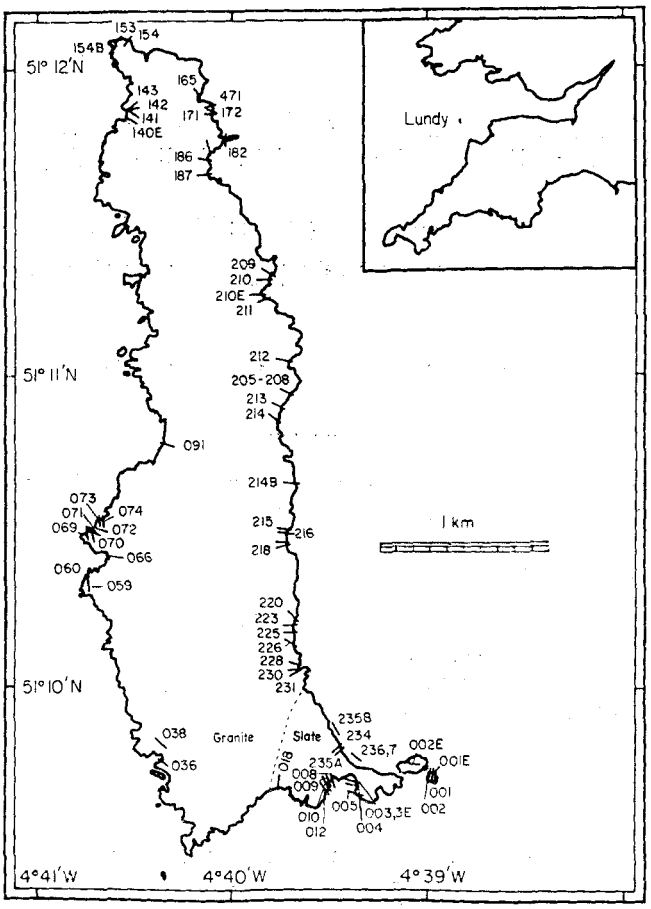


FIG. 1. Map of Lundy showing locations of dykes sampled.

communication) has found magnetic evidence of both small radially-disposed dykes close to the island, and wider dykes (> 10 m) of predominantly NW–SE trend extending further from the island and sometimes showing up on the bathymetry.

2. Collection

Dollar (1941, 1968) reported the presence of about 150 dykes but has not published locations. He found about 91 per cent were dolerites, the rest being types of trachyte.

Sixty-six dykes were sampled, mostly at sea level where exposures are best, and mostly on the east coast which has easier access; thus the chance of the same dyke being sampled on two coasts is small. Locations are shown on Fig. 1; they have also been marked on aerial photographs and photographs have been taken of them and can be made available.

Four cores were taken from all but one dyke, using a portable diamond drill, and orientation was mostly by Sun compass, sometimes using a local sighting point as intermediary.

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3. Magnetic measurements and results

Cylinders 2·5-cm long, 2·2-cm diameter, were cut from the cores and their intensity and direction magnetization was measured using a parastatic magnetometer. They were then progressively demagnetized in alternating fields up to a maximum field of 80 mT (= 800Oe). Susceptibilities were measured using a Bison susceptibility bridge (10^{-4} T, rms; 1000 Hz).

Both natural intensities and susceptibilities show a wide range of values, and their histograms are similar in shape (Fig. 2). There is a strong correlation between these two quantities defined empirically by the equation $\log_{10} M = 1{\cdot}09\chi + 2{\cdot}26$ (M in units of 10^{-4} A m^2 kg^{-1}) with a correlation coefficient of 0·89. It might seem that the samples only differ—as far as their magnetic properties are concerned—in the amount of magnetic material present, but this cannot be entirely true because the low intensity samples were considerably less stable than the stronger ones (Fig. 2). On demagnetization the intensities decreased rapidly, as is often the case with dykes from this province, (Ade-Hall *et al.* 1972).

There is no general agreement how the directions of magnetization of individual samples—which change with the value of the demagnetizing field—should be combined

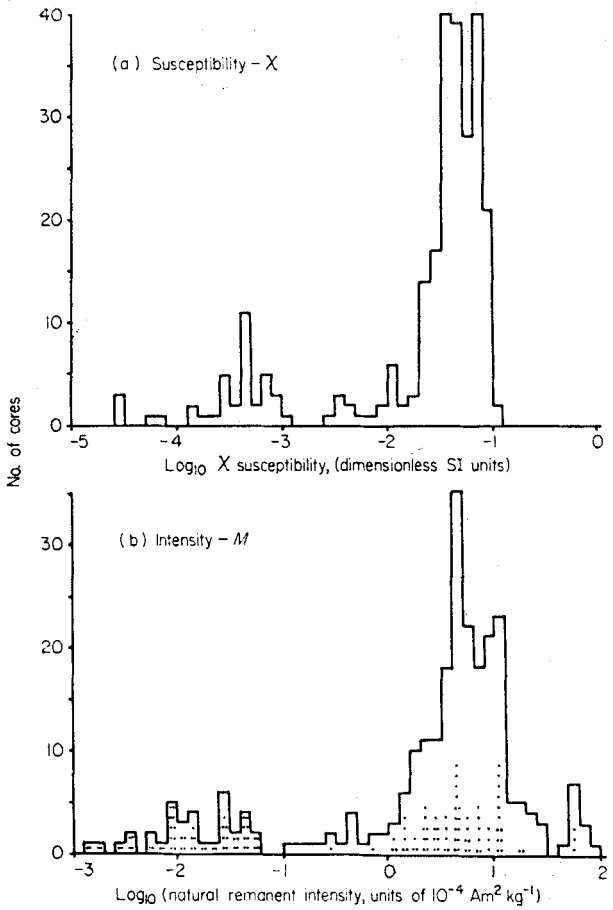


FIG. 2. Histograms of (a) susceptibilities, (b) natural remanent intensities. On (b) cores that were not included in calculating the means are indicated by a dot; if the remaining cores of the same dyke were also rejected a second dot has been added.

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to give the mean direction of the body from which they came; this is because the factors that cause variation in direction are not fully understood. Therefore two methods were tried with the Lundy data.

The first method is that used as standard by this laboratory (Dagley & Ade-Hall 1970). Basically it consists of finding the tightest grouping of the directions of all the samples of a single dyke, regardless of the demagnetizing field used, and the grouping is expressed as a mean direction and an α_{95}. Dykes were then rejected if they had $\alpha_{95} > 10°$, a value chosen because it formed a natural gap in the distribution of α_{95}'s. Six of the seven most unstable dykes were of the low-intensity trachytic kind. Results are given in Table 1 and Fig. 3.

Fifty-two dykes gave a stable direction according to the above criterion. Of these four were classified as intermediate, having virtual pole positions more than 40° from the geographic pole (Wilson *et al.* 1972); four were reversely magnetized and the remaining 44 were normal. The normal dykes give an overall mean direction of N 9·4 °E, +68·9° (down) with $\alpha_{95} = 3·54°$, which is indistinguishable from the centred axial dipole field direction of +68° expected at this latitude of 51°1 °N. The 44 reversed dykes have mean direction N 183·2 °E, –62·1° with $\alpha_{95} = 1·60°$ which is significantly shallower than expected. With so few normal dykes it is felt that the apparent asymmetry of direction of normal and reversed directions is not significant, and the mean direction of the combined normal and reversed dykes is N 3·6 °E, +62·7° (down) with $\alpha_{95} = 1·5°$, which also is shallower than expected.

The second method of selecting stably-magnetized dykes (used for the first time in this paper, and termed the Briden–McCormack method) is based on the stability criterion of Briden (1972). In this the ' best ' direction of each core is chosen as that

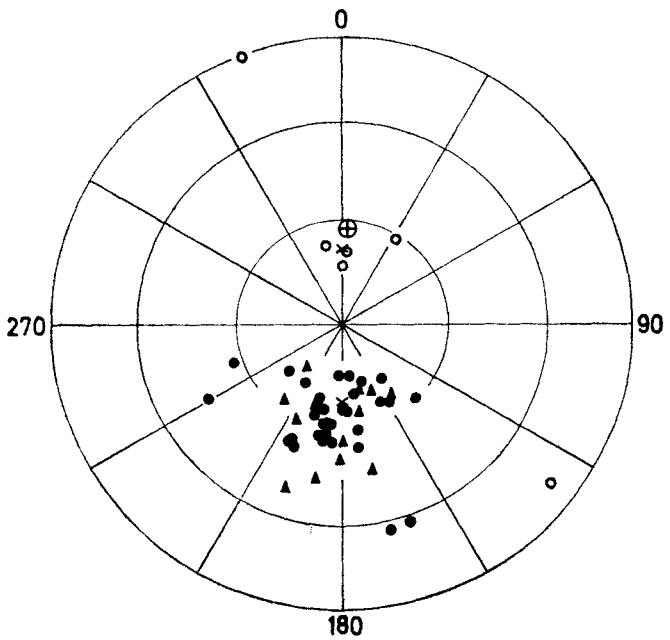


FIG. 3. Mean directions of dykes with $\alpha_{95} \leqslant 10°$ calculated according to the method of Dagley & Ade-Hall (1970). Equal area stereographic projection; Open circles denote positive (downward) inclination—lower hemisphere; full circles denote negative (upward) inclination—upper hemisphere; X, denotes centred axial dipole field direction; ⊕, denotes overall mean direction.

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Table 1

Palaeomagnetic directions

	Code	Dagley & Ade-Hall N	D (°ETN)	I (°down)	α (°)	Code	N	D (°ETN)	I (°down)	α (°)
A14T	001	3	168·9	−46·7	4·9	182	3	171·0	−70·2	5·2
	001E	2	179·3	−63·4	31·5	186	3(5)	228·5	−70·2	4·1
	002	3	205·0	−59·4	6·3	187	3	3·0	69·7	2·8*N*
	002E	2	217·0	−62·9	1·0	205	4(5)	167·0	−27·3	6·1
	003	2	195·3	−62·0	16·4	206	4	179·9	−66·0	2·5
	003E	3	198·9	−39·8	3·2	207	3(3)	177·3	−65·3	2·0
	004	2	179·7	−55·8	3·5	208	4	154·0	−65·6	3·3
	005	0	—	—	—	209	3	172·3	−60·0	2·5
	008	3	189·6	−44·5	3·8			(180)	(−53)	
	009	0	—	—	—	210	4	196·6	−63·1	6·1
	010	4	181·1	−51·0	7·7			(211)	(−58)	
	012	2	198·2	−65·3	3·2	210E	3	347·9	67·1	4·1*N*
	018	4	218·6	−74·4	2·8	211	3	155·8	−50·6	12·8
	036	2	135·4	−60·4	3·2	212	0	—	—	—
	038	4	204·4	−53·0	1·9	213	2	191·4	−61·3	3·7
	059	4	250·2	−57·4	5·6*I*	214	0	—	—	—
	060	2	161·0	−28·0	8·4	214B	2(3)	191·9	−65·2	3·4
	066	0	—	—	—	215	2(2)	127·5	10·5	9·6*I*
	069	4	186·2	−61·2	2·6	216	2(3)	339·7	1·4	3·6*I*
	070	4	196·7	−67·4	1·5	218	4	172·5	−54·6	4·7
	071	4	172·3	−75·5	1·6	220	4	359·8	73·7	3·9*N*
	072	4	183·6	−75·6	9·7	223	3	148·7	−64·2	5·6
	073	4	189·8	−56·2	4·2	225	4	137·7	−70·7	13·4
	074	3(3)	188·4	−58·5	7·6	226	3	185·3	−56·1	6·2
	091	3	240·6	−46·4	3·2*I*	228	2(2)	157·2	−31·7	11·5
	140E	3	211·7	−70·2	5·2	230	4	192·3	−58·0	6·3
	141	2	187·2	−61·4	4·6	231	5(5)	203·5	−54·5	4·4
			(187)	(−57)						
	142	one core only				234	4	145·4	−65·8	6·3
	143	5(5)	201·4	−52·4	4·9			(186)	(−66)	
	153	4	32·2	61·5	6·9*N*	235A	0	—	—	—
	154	2(5)	191·0	−61·1	1·6	235B	4	169·1	−64·3	1·7
	154E	3	143·9	−71·5	5·3	236	3	156·7	−69·5	7·9
	165	2	197·6	−68·7	6·3	237	3(6)	183·4	−70·2	11·7
	171	3(3)	162·7	−73·2	7·2	471	4	166·4	−70·5	5·3
	172	0	—	—	—					

Dykes 001–018 and 234–471 intrude slate; the remainder intrude granite.
N no of cores used to find mean; four cores collected for all, except where indicated.
D declination—degrees east of true north.
I inclination—positive downwards: all dykes reversed except as indicated intermediate, *I*, or normal, *N*.
α angle of 95 per cent confidence.

Blundell's result where applicable in parenthesis.

direction which shows least vector change during progressive demagnetization. The next step is to compare this ' best ' direction with the mean direction of the other cores of the dyke; this is done for each core in turn and at the end of the round cores are rejected if their direction is more than two circular standard deviations from the mean of the others. This procedure is repeated with the reduced number of samples until one of several possibilities ensues: no more samples are rejected; only two samples are left; the standard deviation of the group becomes 7·5° or less (arbitarily considered to be a reasonably tight grouping). The purpose of this procedure is to reject as

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objectively as possible any samples that are widely divergent in direction from the rest. It is not fully satisfactory, for if directions are approximately equally-spaced along a great circle it reduces three such samples (say) to a single one. The problem arises because the distribution is not even approximately circularly Gaussian, a situation quite likely to occur when sample numbers are small.

When the acceptable samples have been defined as above the mean direction and α_{95} are calculated. As it can still happen that these samples have widely divergent directions a cut-off has still to be applied. This was chosen as $\alpha_{95} > 20°$, a value twice that of the previous method because this procedure gives more scattered groupings since it does not directly minimize α_{95}.

This method gave mean dyke directions considerably different from the first method for some dykes; in 13 cases a difference of more than 10° was found, though in some of these α_{95} was also greater than 10°. Forty-one dykes were accepted (compared to 52 by the first method), comprising two intermediate, four normal and 35 reversed, and the mean direction of the combined normal and reversed dykes was N 5·9 °E, +62·9°, $\alpha_{95} = 2·49°$. This is not significantly different from the value calculated by the first method, which accordingly is used throughout this paper, to be consistent with other British Tertiary directions published by this laboratory.

As the K–Ar dates were not accurate enough to distinguish with confidence between the times of intrusion of different dykes—as will be described below—attempts were made to see if any differences could be distinguished palaeomagnetically. The dykes cutting the granite are obviously younger, but those cutting the slate could be older. However, no significant difference was found between these two groups of dykes (of the dykes cutting the granite 35 were acceptable and gave a direction N 4·0 °E, +63·2°, $\alpha_{95} = 1·7°$; the corresponding figures for the 13 dykes cutting the slate were N 2·7 °E, +61·3°, 3·1°). An attempt to distinguish between the wide NW–SE-trending dykes and the narrower, more local, radial swarm was frustrated because the strikes and widths of dykes on the island were very variable, some even bending through a right-angle in a distance of less than a metre.

In the absence of any significant groupings a pole position was calculated for the 48 dykes that gave well-defined normal or reversed directions according to the first method. If calculated from the mean direction the mean pole position is 82·6 °N, 155·0 °E ($dp = 1·8°$, $dm = 2·3°$); it is 84·0 °N, 150·7 °E ($\alpha_{95} = 1·90°$) if calculated from the individual pole positions of each dyke. The difference is not significant and the former is preferred as the scatter is probably due to collection and measuring errors plus secular variation, rather than motion of the mean axial dipole which was likely to have been small in the time involved (see next section). Neither value includes the present pole within its cone of confidence. The ' common site longitude ' pole (Wilson 1972) is 82·6 °N, 159·4 °E of site, and shows the right-handed far-sidedness discussed by Wilson.

The mean direction of magnetization given in this paper is more precise than that obtained by Blundell (1957), which was N 197 °E, −59°, $\alpha_{95} = 7°$ and the circles of confidence just overlap, the poles being 7·5° apart.

4. K–Ar dates

Thin sections were made of all cores and were examined under the microscope. All showed some degree of alteration. Only the freshest were selected for dating, in which the feldspars were little altered, displaying only cracking and/or a small amount of sericite.

The degree of alteration is indicated in Table 2, according to the scale of Mussett *et al.* (1973) (based chiefly on the degree of alteration of feldspars, one indicating fresh feldspars, five completely altered ones). This is intended only as a guide as experience

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Table 2

Ages of Lundy dykes

Sample	Alteration*	Potassium (% K)	$A^{40}_{rad}/g \times 10^6$ ccstp	$A^{40}_{rad}/A^{40}_{total}$ (%)	Age (My)
T002E	2a	0·555	1·22	54	54·3±1·0
T005	1a−	5·62	11·5	76	50·9±1·0
T006	2a	4·74	9·39	91	50·7±1·0
T006	2a	4·74	9·25	93	48·3±0·7
T141	2a−	4·13	7·42	97	44·6±1·0
T141	2a−	4·13	8·54	88	45·4±0·8
T211	3a+	2·50	5·02	53	49·7±0·8
T213	2a	6·23	12·7	72	50·2±0·7

$\lambda_\beta = 5{\cdot}30 \times 10^{-11}\, a^{-1}$, $\lambda_\varepsilon = 0{\cdot}585 \times 10^{-11}\, a^{-1}$, $K^{40}/K = 1{\cdot}19 \times 10^{-4}$ (atom ratio)

* following scheme of Mussett *et al.* (1973).

has shown (Mussett *et al.* 1973) that it is not a very reliable indicator of argon retention in fine-grained volcanic rocks.

The dated samples divide into two groups: T002E is a basalt, with quite fresh feldspars, altered mafics and no glass. The other samples have large amounts of potassium which is attributed to the incorporation of large amounts of slate; they contain no olivine or pyroxene but do contain devitrified glass.

Argon was extracted by fusion and analysed using a Vacuum Generators Micromass 6 mass spectrometer; potassium was determined using an EEL 100 flame photometer, following the general procedure given in Purdy *et al.* (1972). Results are given in Table 2.

The dates span an apparent interval of 54–45 My ago. However, these values need to be used with caution as experience has shown that lavas of the British Tertiary often give erroneous ages, as proved by inconsistency with the stratigraphy. This is probably due to argon loss, or perhaps initial argon (Mellor & Mussett 1975). Less is known about the reliability of dyke ages because of the poor stratigraphic control, but they are broadly similar in petrology to the lavas and so are suspect to the same degree. The most reliable is probably that of the basalt T002E, because it contains no glass.

Some limits on the dyke ages can be set in other ways. The granite has been reliably dated by both the K–Ar and Rb–Sr methods (Miller & Fitch 1962; Dodson & Long 1962) and gives a mean age of 52·7±1 My. This provides an upper limit to the age of all dykes except those that cut the slate. T002E is one of these latter dykes and gave an age of 54·3±1 My which is not significantly different from that of the granite.

The south-west of England, including Lundy, is traversed by numerous NW–SE trending faults that have a dextral component, and some of these have offset both the dykes on Lundy and the single dyke in Lee Bay on the mainland which is thought also to be part of the swarm. The Sticklepath–Lustleigh Fault is part of this system, and it seems likely from the gravity interpretation that it has displaced the basic mass postulated beneath the Lundy area (Brooks & Thompson 1973). In South Devon, Oligocene clays are affected by the faulting, and it is thought that the main movement occurred about this time, though some earlier movement is not precluded (Shearman 1967). This gives a lower limit of about 35 My, which however is considerably less than any of the dates obtained for the dykes.

It is tentatively concluded that the dykes were intruded in the approximate period 54–45 My ago, but no time sequence can be given for the swarm.

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5. Discussion and conclusions

A feature of the results is the small proportion of normally magnetized dykes, four out of the 48 dykes that gave well-defined normal or reversed directions. A minority of normally magnetized dykes has also been found in the swarms of Mull (27 per cent of 154 dykes; Ade-Hall *et al.* 1972), the Vaternish dykes of north Skye (19 per cent of 73 dykes; Wilson *et al.* 1974), and Arran (18 per cent of 442 dykes; unpublished work). The proportion in Lundy at 8 per cent is less than any of these, though with the small number of dykes the statistical error could be quite large. Interest arises because some at least of these other swarms are up to several million years older than Lundy so that the roughly similar proportions of normal and reversed dykes cannot be because the swarms were closely contemporaneous. The Mull dykes are about 60 My old (Mussett *et al.* 1973), the Arran swarm pre-dates the Holy Island sill which has been dated at 55–56 My (Macintyre 1973), and the Skye swarm pre-dates the Red Hills centre whose age is about 53 My (Long 1964; Moorbath & Bell 1965; Kelly 1973), whereas the bulk of the Lundy dykes, including all four normal ones, post-date the granite (53 My), and as shown above, may be considerably younger. Thus the predominantly reversed direction of magnetization in the British Tertiary igneous province cannot be entirely due to formation in a period comparable to a polarity epoch, as suggested by Wilson (1970).

The other conclusion of this paper is that the mean direction of magnetization of the Lundy dykes, aged about 54–45 My, is N 3·6 °E, +62·7° (down), with $\alpha_{95} = 1{\cdot}5°$. This yields a pole position of 82·6 °N, 155·0 °E with $dp = 1{\cdot}8°$, $dm = 2{\cdot}3°$, which differs significantly from the present geographic pole, and displays the right-handed, far-sidedness discussed by Wilson (1972).

The possible implications of those observations will be discussed, together with other results for the province, in a future paper.

Acknowledgments

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